



Lecture 6: PN junction under bias

ECE 340 – Electronic Circuits I Sec. 1

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09/22/2026

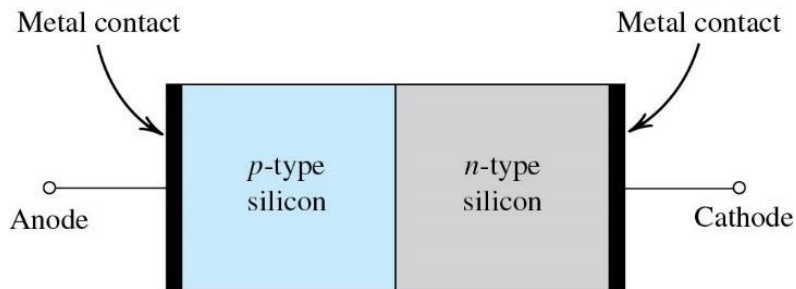
(textbook sections 3.3-3.5)



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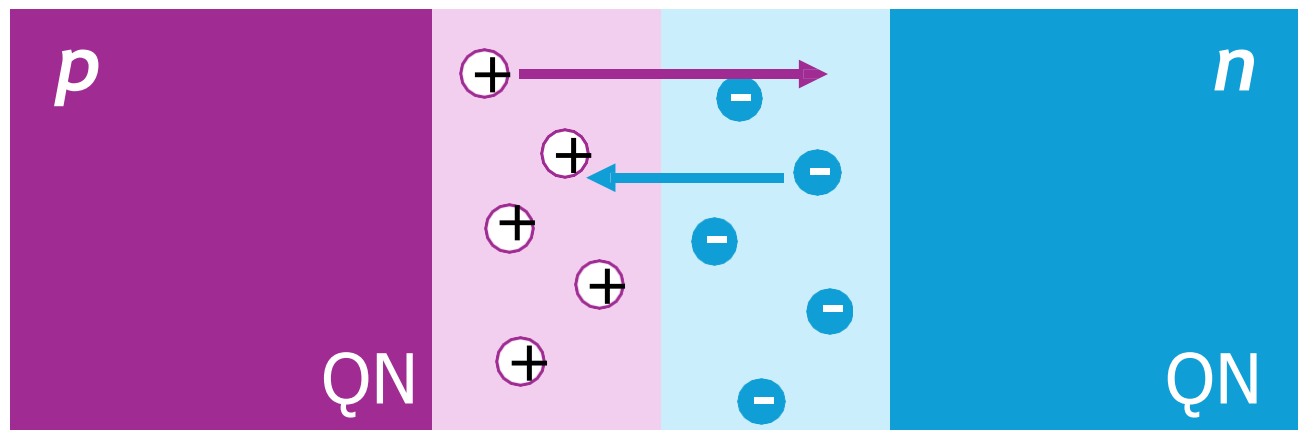
Previously...

pn-junction (no applied voltage)



The holes recombine with some **majority electron carriers** on the **right**, causing a *depletion of free electrons* close to the junction

The electrons recombine with some **majority hole carriers** on the **left**, causing a *depletion of holes* close to the junction



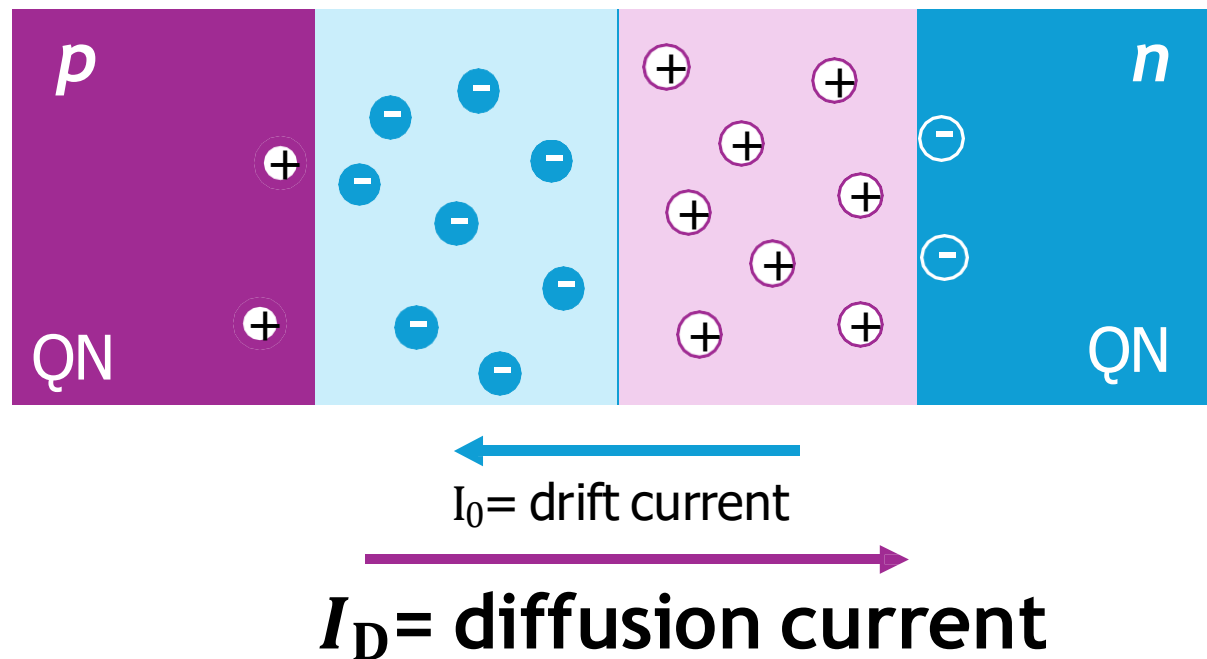
$I_D =$ diffusion current

Previously...

Why does it stabilize?

Say $I_D > I_S$. More carriers diffuse to opposite sides

V_0 & Electric field increases, causing a slowdown in I_D



Eventually, $I_D = I_S$ (equilibrium)

Depletion region width

$$N_A = N_D:$$

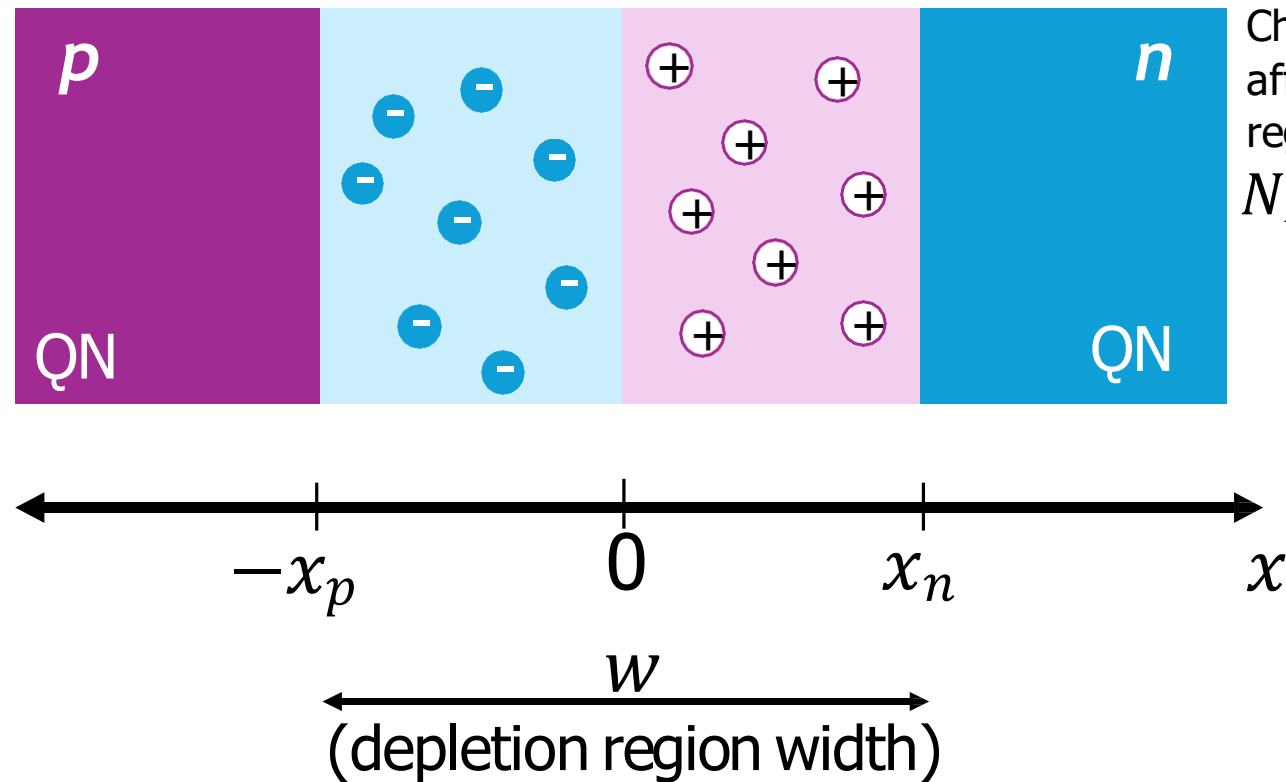
Within the depletion region, NO charges are mobile (but there is net charge due to fixed dopants)

Amount of charge is the same:

$$qAx_nN_D = qAx_pN_A,$$

where A = cross-sectional area

Charge density affects depletion region width:
 $N_Ax_p = N_Dx_n$



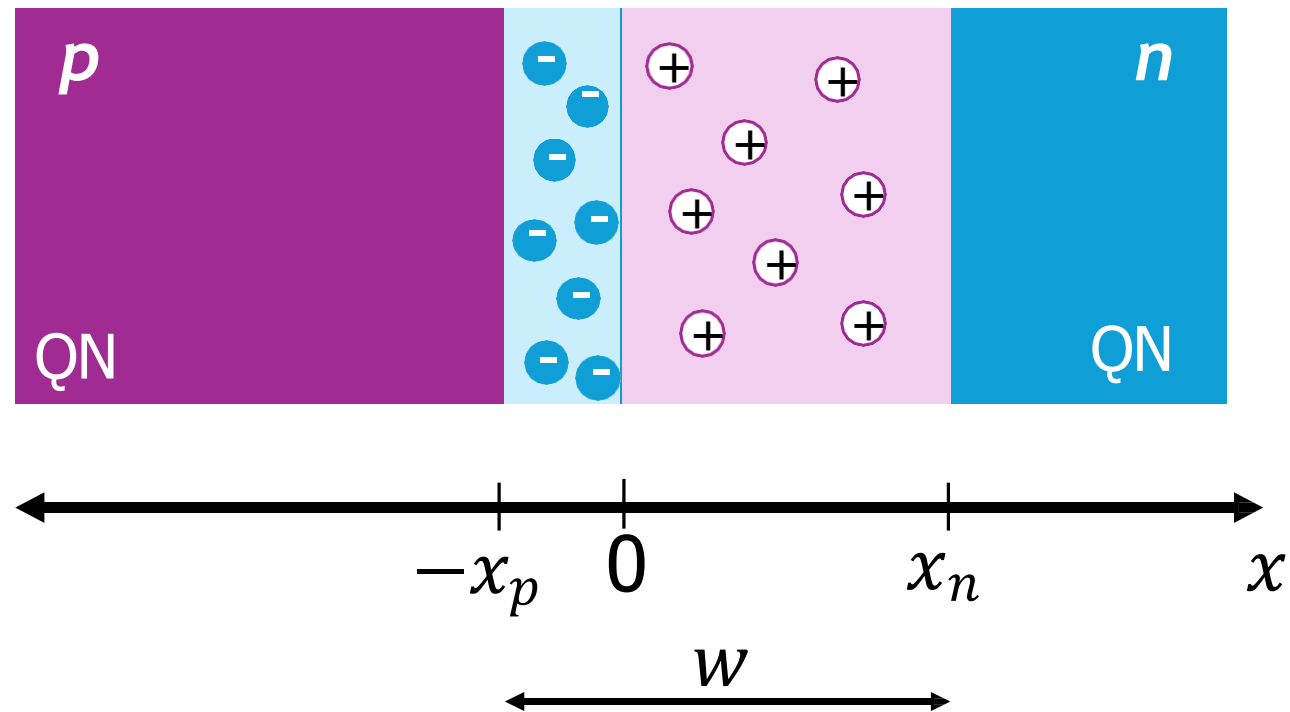
Depletion region width

We can either have $N_A \gg N_D$ or $N_A \ll N_D$ (due to the way we fabricate)

$$N_A \gg N_D:$$

Charge density
affects depletion
region width:

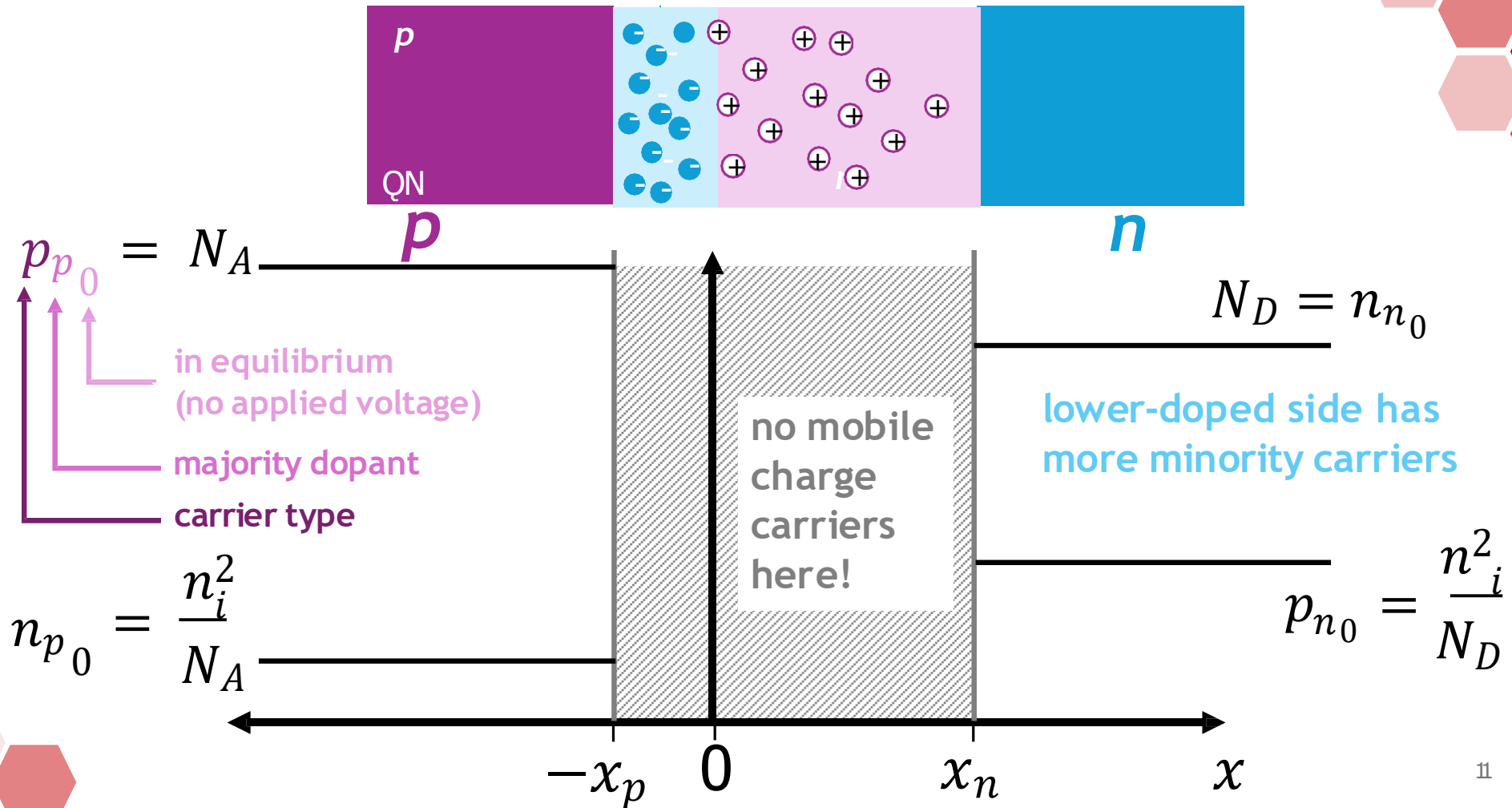
$$N_A x_p = N_D x_n$$



Visualizing carrier concentration

$$N_A > N_D:$$

$$n^2 = np$$



Depletion region width

$$w = x_p + x_n = \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) V_0}$$

where $\epsilon_s = \epsilon_r \epsilon_0$, the electrical permittivity of Si.

$$\epsilon_0 = 8.85 * 10^{-14} \text{F/cm (or } \epsilon_0 = 8.85 * 10^{-12} \text{F/m)}$$

$$\epsilon_r(\text{Si}) = 11.7$$

V_0 = built-in voltage,

$$V_T = kT/q = 25.9 \text{ mV}$$

w is typically 0.1 μm - 1 μm ; depends on N_A and N_D (specifically, the smaller of the two)

$$V_0 = V_T \ln \left(\frac{N_A N_D}{(n_i)^2} \right)$$

For each region:

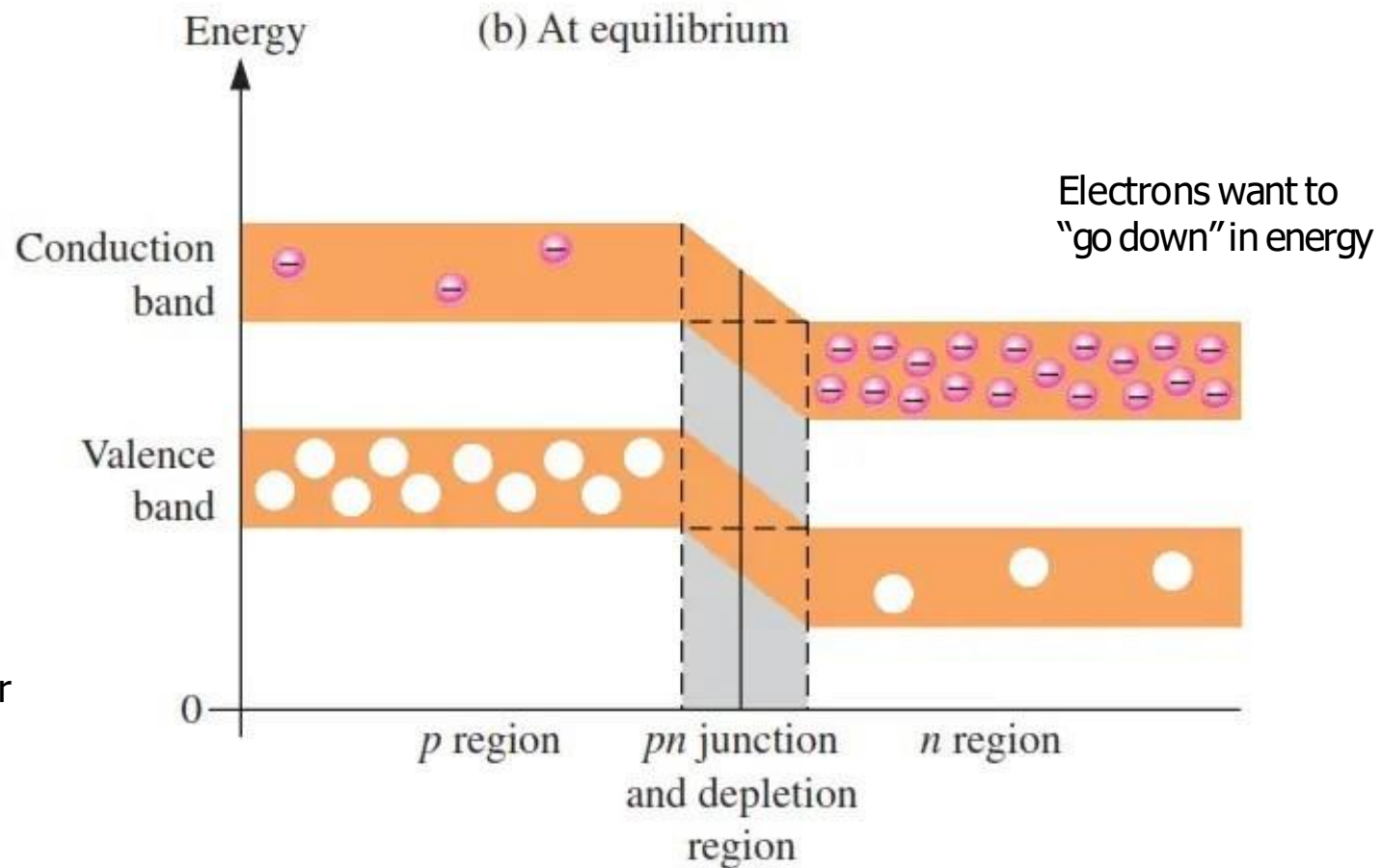
$$N_A x_p = N_D x_n$$

$$x_n = w \frac{N_A}{N_A + N_D}$$

$$x_p = w \frac{N_D}{N_A + N_D}$$

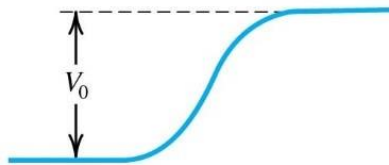
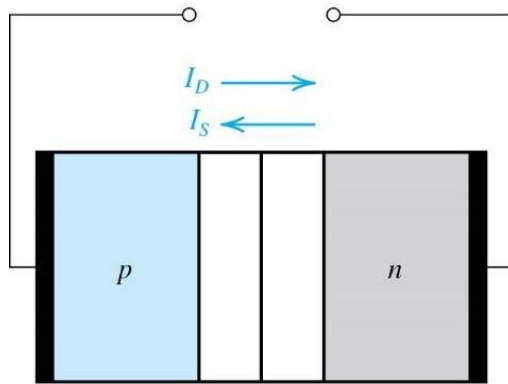
Bonus: *pn* energy band diagrams

Holes want to
“float” to higher
energy
(e.g., electrons
settling into lower
energy orbitals)



***pn* junction under bias**

drift (I_S) & diffusion (I_D) currents play a role



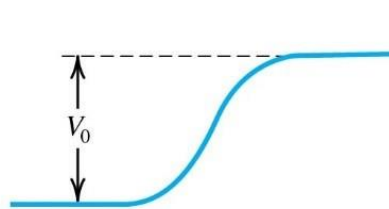
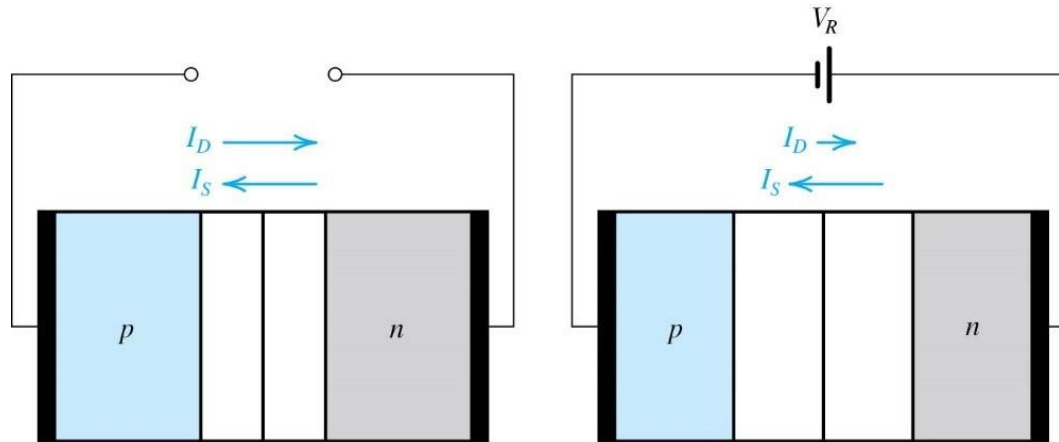
(a) Open-circuit
(equilibrium)

$$I_D = I_S$$

***pn* junction under bias**

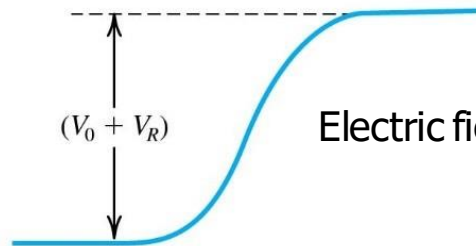
drift (I_S) & diffusion (I_D) currents play a role

Built-in electric field



(a) Open-circuit
(equilibrium)

$$I_D = I_S$$



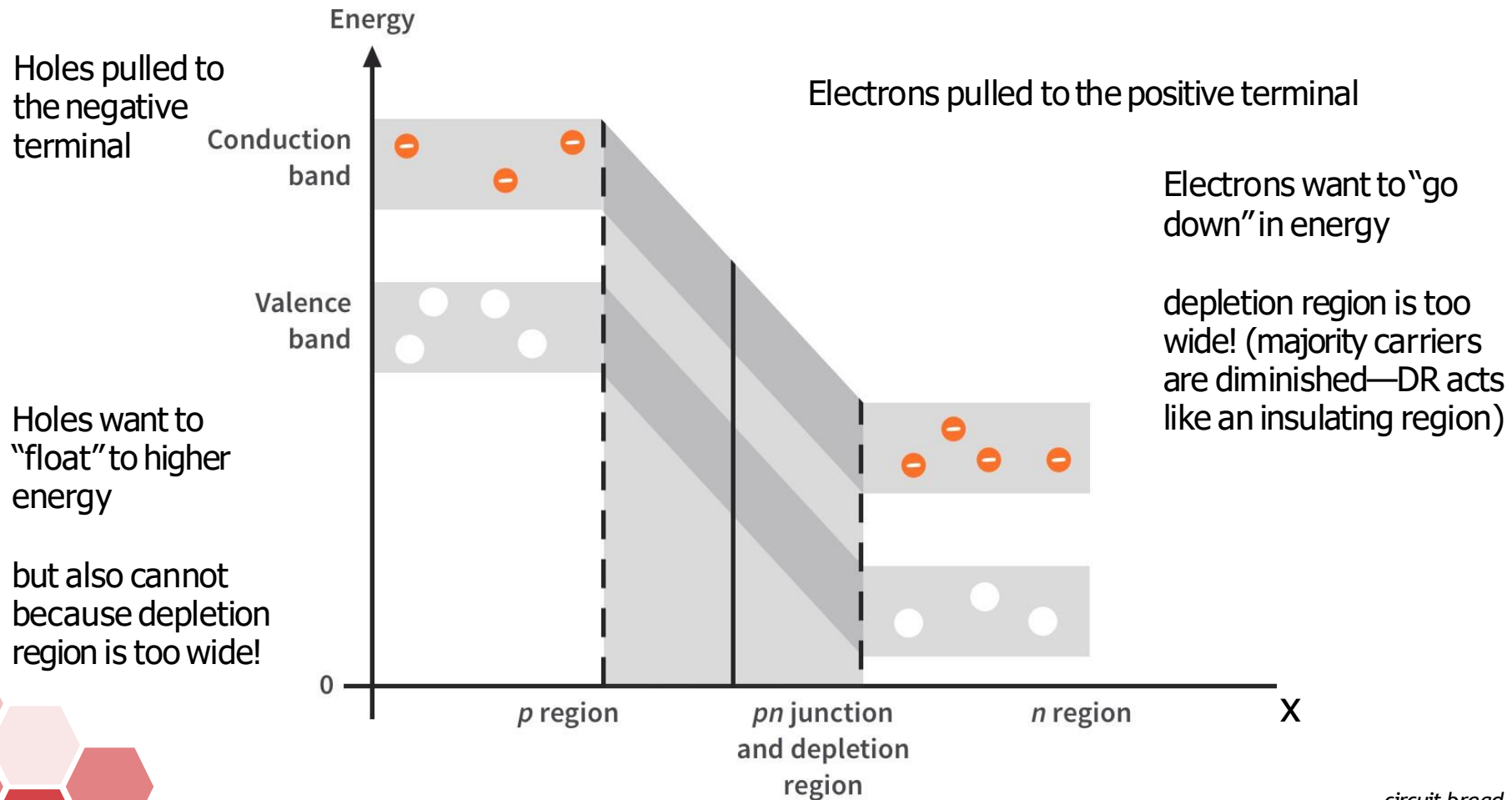
(b) Reverse bias

$$I_D < I_S$$

Electric field across the junction increases

(typically, I_S is extremely small unless V_R exceeds the "breakdown voltage" – more on that to come...)

Reverse-biased pn junction - energy



Reverse-biased *pn*-junction width

$$w = x_p + x_n = \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (V_0 + V_R)}$$

We gain a term

where $\epsilon_s = \epsilon_r \epsilon_0$, the electrical permittivity of Si.

$\epsilon_0 = 8.85 * 10^{-14} \text{F/cm}$ (or $\epsilon_0 = 8.85 * 10^{-12} \text{F/m}$)

$\epsilon_r(\text{Si}) = 11.7$

V_0 = built-in voltage,

$V_T = kT/q = 25.9 \text{mV}$

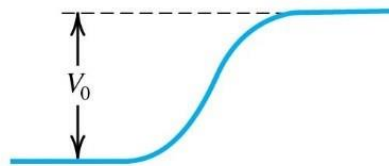
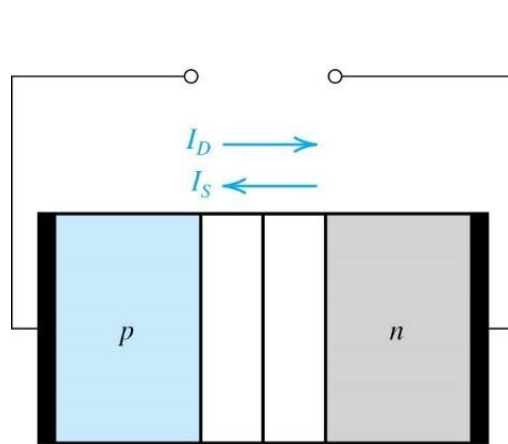
$$Q_J = qA \left(\frac{N_A N_D}{N_A + N_D} \right) w = A \sqrt{2\epsilon_s q \left(\frac{N_A N_D}{N_A + N_D} \right) (V_0 + V_R)}$$

$Q_J, w \uparrow \text{ as } V_R \uparrow$

***pn* junction under bias**

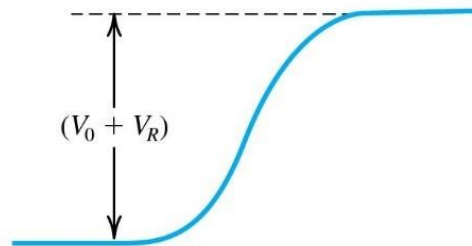
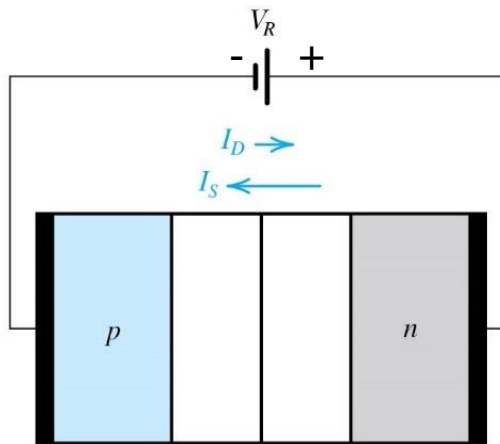
drift (I_S) & diffusion (I_D) currents play a role

Built-in electric field



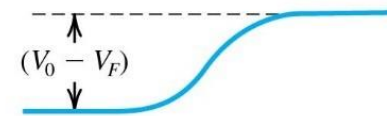
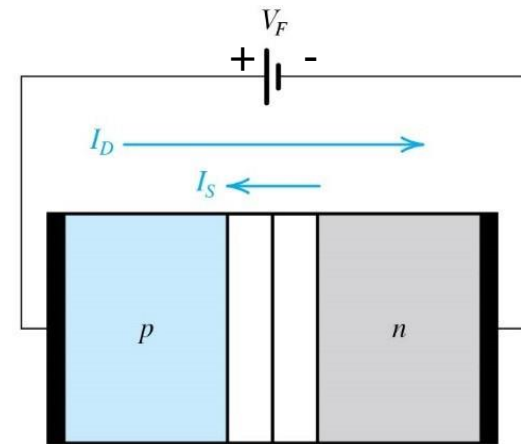
(a) Open-circuit
(equilibrium)

$$I_D = I_S$$



(b) Reverse bias

$$I_D \ll I_S$$

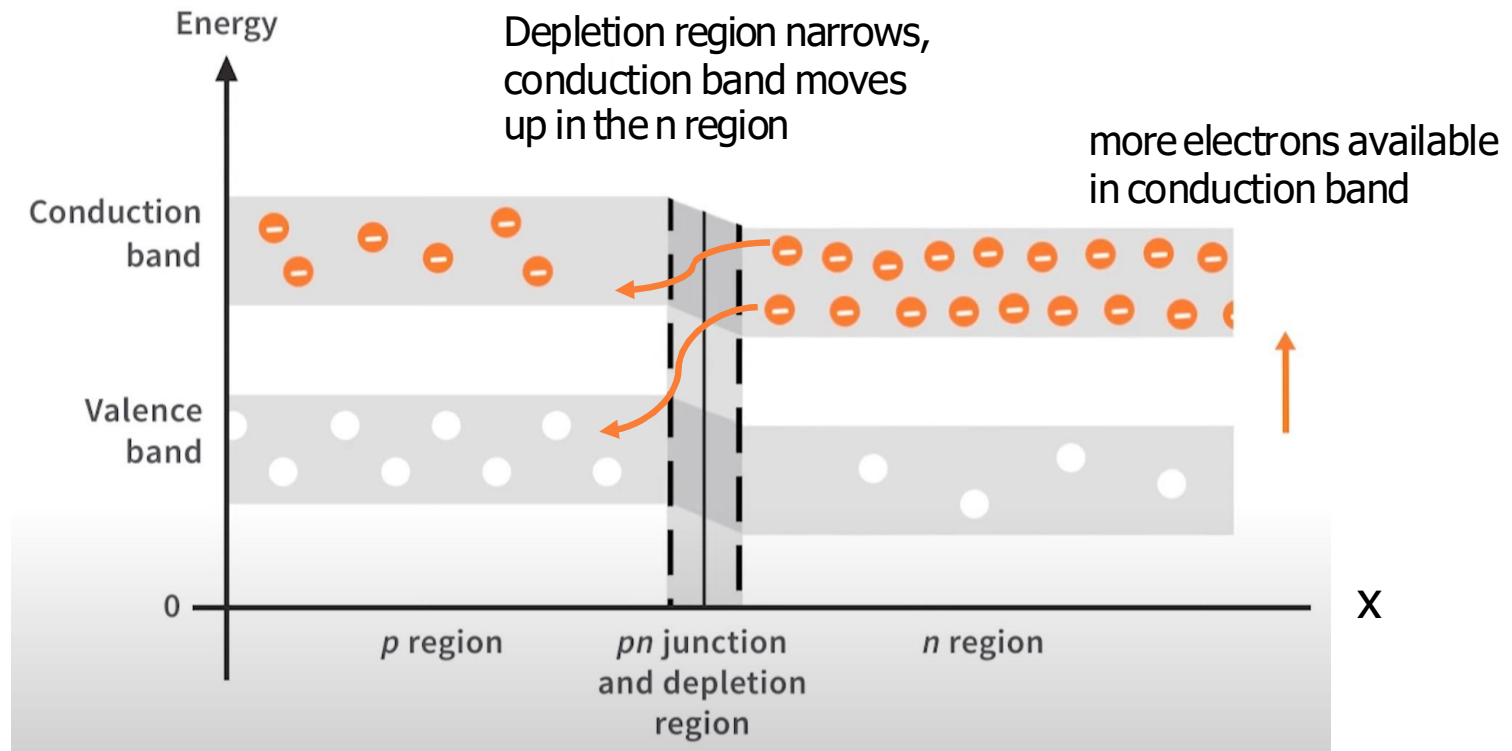


(c) Forward bias

$$I_D \gg I_S$$

Forward-biased pn junction - energy

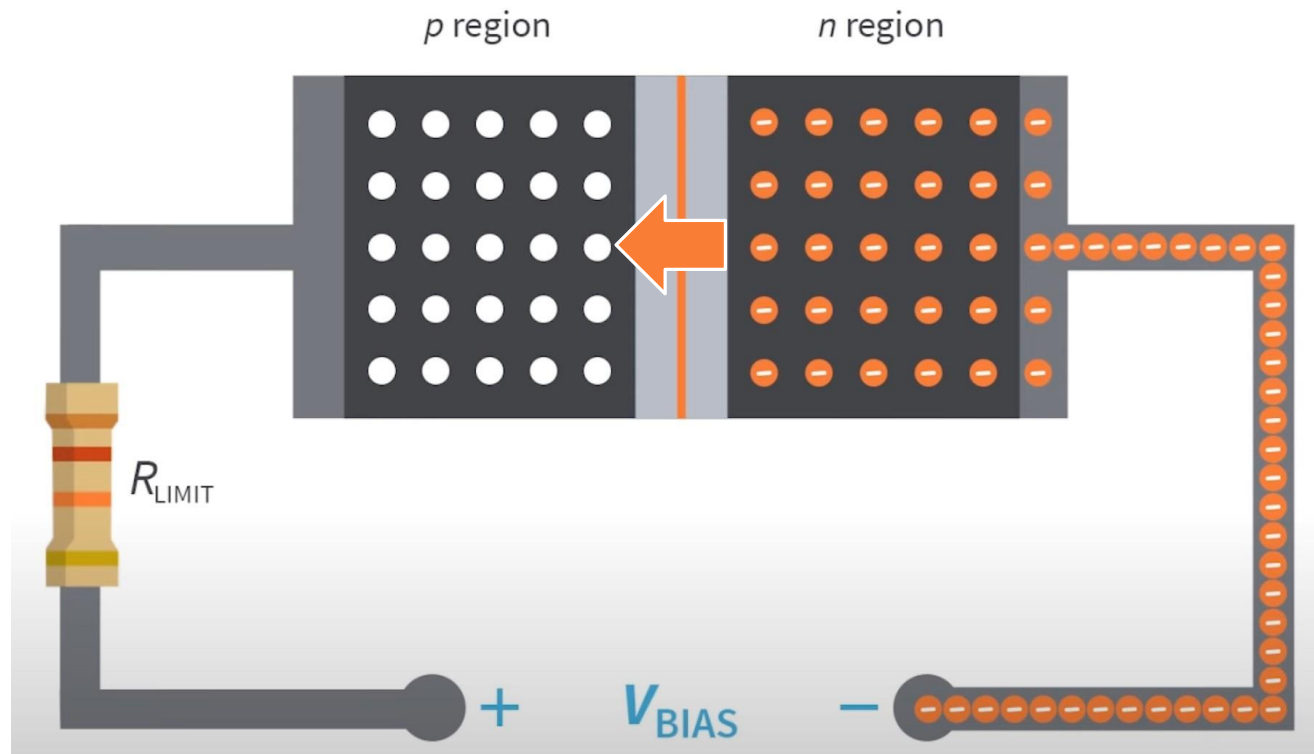
Electrons crossing the depletion region isn't free! They lose an amount of energy equal to the barrier potential.



So the voltage dropped across the junction is 0.7 V

* There's an additional drop due to internal resistance
("dynamic resistance", usually negligible)

pn-junction under forward bias



<https://www.circuitbread.com/tutorials/how-does-a-diode-work-part-1-the-pn-junction>

circuitbread

Forward-biased pn-junction width

$I = I_D - I_S \cong I_D$ – majority carrier current, depends on barrier height
(barrier is reduced)

$$w = x_p + x_n = \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (V_0 - V_F)}$$

$$Q_J = qA \left(\frac{N_A N_D}{N_A + N_D} \right) w = A \sqrt{2\epsilon_s q \left(\frac{N_A N_D}{N_A + N_D} \right) (V_0 - V_F)}$$

$Q_J, w \downarrow$ as $V_F \uparrow$

Generalized relationship

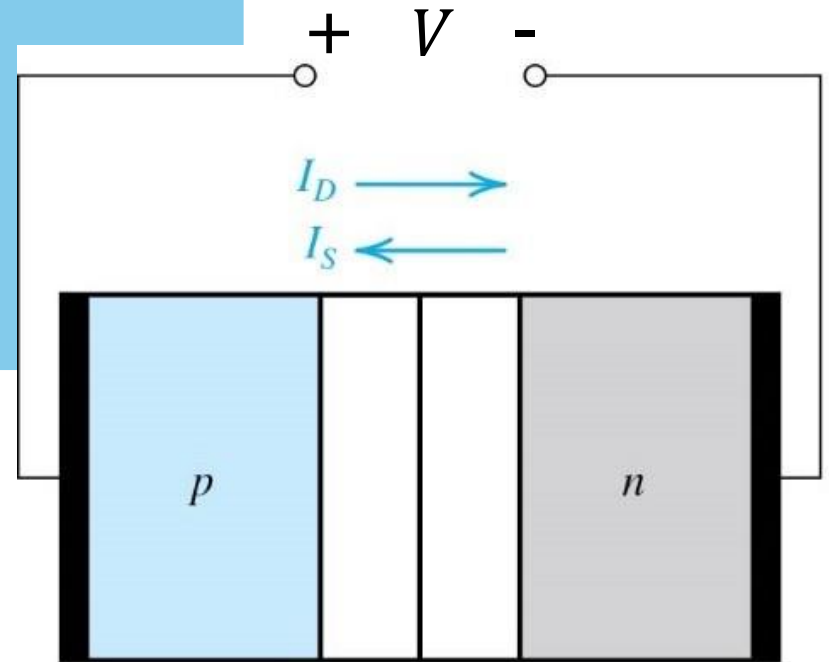
In general: $w \propto \sqrt{(V_o - V)}$

$$Q_J \propto \sqrt{(V_o - V)}$$

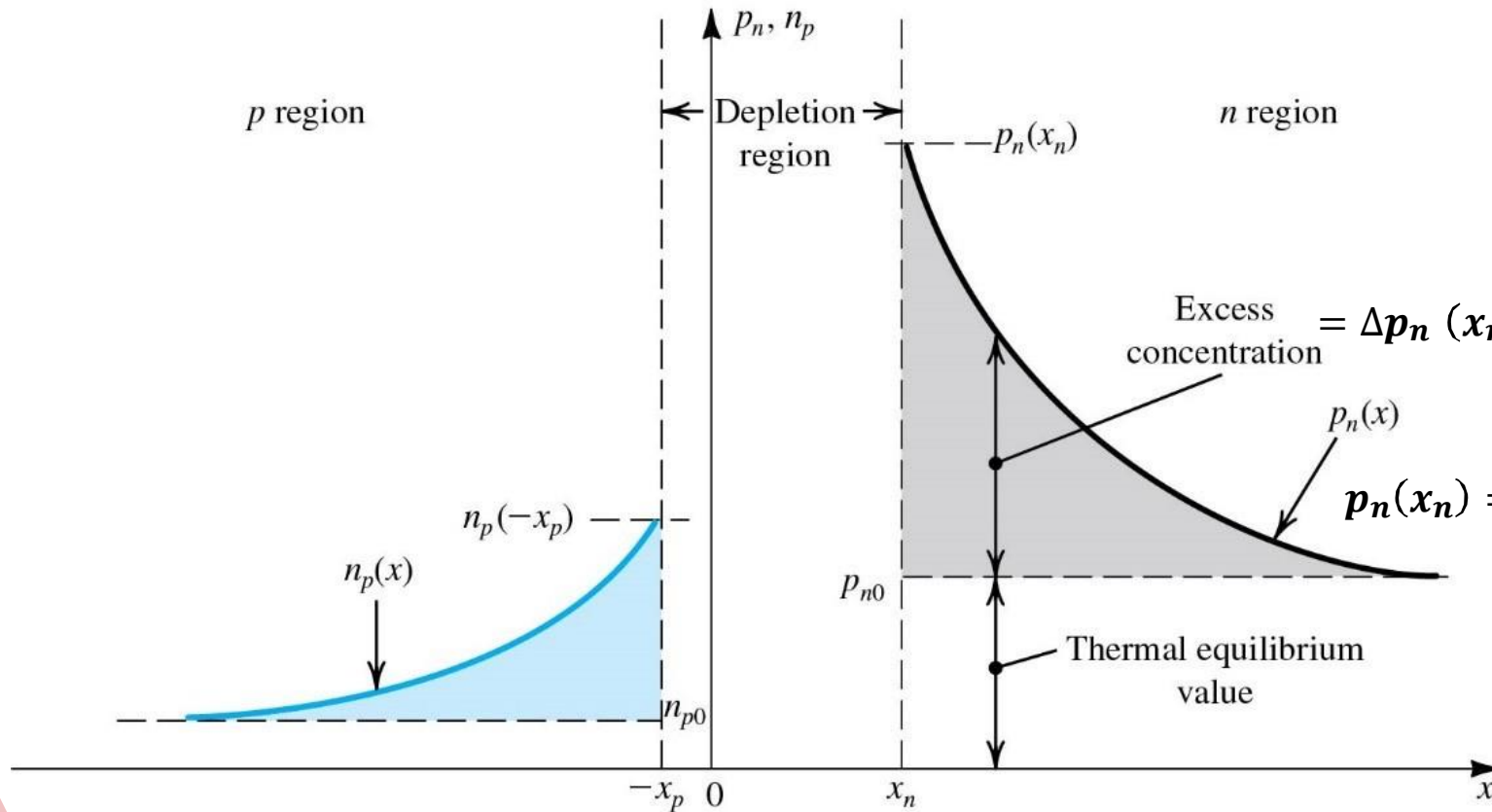
where if

$V > 0$, it is forward biased

$V < 0$, it is reverse biased



Forward bias carrier concentration



Excess hole concentration (compared to equilibrium level p_{n0})

$$\begin{aligned}
 &= \Delta p_n(x_n) \\
 &= p_{n0} e^{\frac{V}{V_T}} - p_{n0} \\
 &= p_{n0} (e^{\frac{V}{V_T}} - 1)
 \end{aligned}$$

$$p_n(x_n) = p_{n0} e^{\frac{V}{V_T}}$$

$$V_T = \frac{k_B T}{q}$$